# Sahayog HRMS

AWS Cloud Infrastructure & Technical Architecture Report

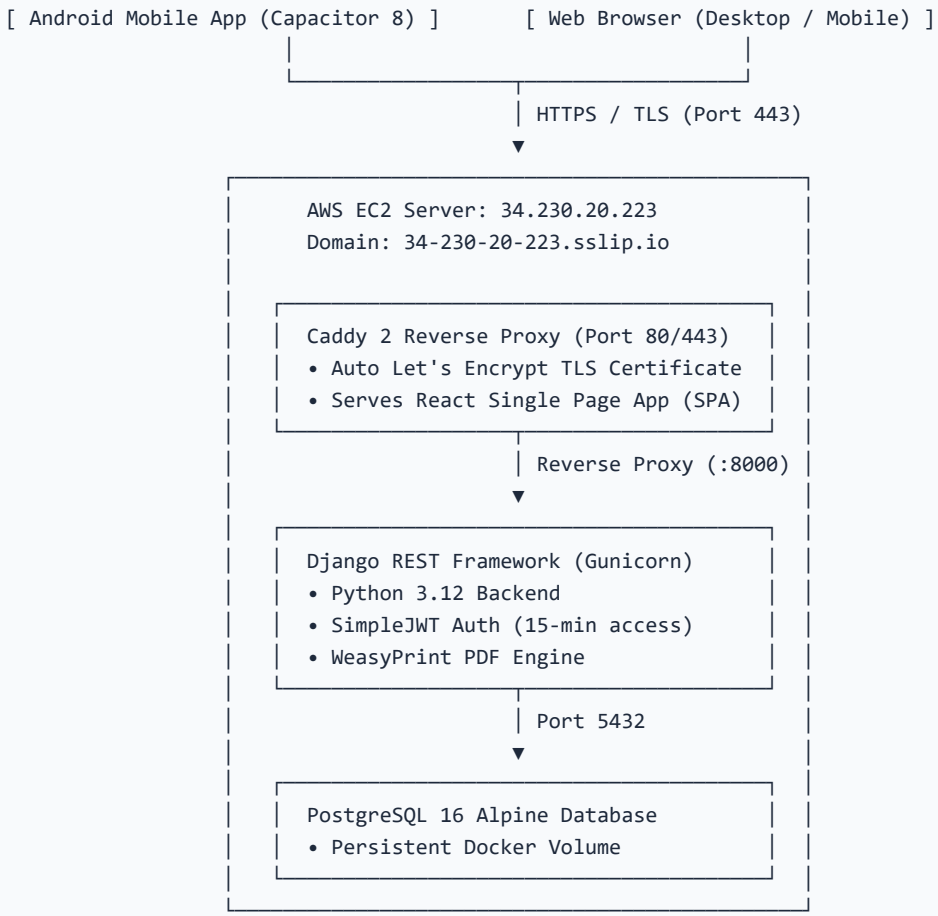
● PRODUCTION LIVE

**Date:** October 22, 2026
**Target Cost:** ₹0 / $0 (AWS Free Tier)

## 1. Executive Summary

The Sahayog HR Management System is hosted in a high-efficiency, fully containerized production architecture deployed on **Amazon Web Services (AWS)**. The system is provisioned via **AWS CloudFormation** and optimized for the **AWS Free Tier**, enabling zero operational hosting costs while delivering full HTTPS TLS encryption, automatic subdomain routing, persistent database backups, and high availability for both web users and the native Android mobile app.

## 2. End-to-End System Architecture




## 3. AWS Infrastructure Specifications

| Component | AWS Specification | Description & Purpose |
|---|---|---|
| **Cloud Compute** | `EC2 t3.micro` | 2 vCPU burstable, 1 GB RAM, AWS Free Tier eligible. |
| **Instance ID** | `i-0326b8a57385679b1` | Active virtual machine instance in AWS EC2 console. |

| Component | AWS Specification | Description & Purpose |
| --- | --- | --- |
| **Region & AZ** | `us-east-1 (N. Virginia)` / `us-east-1c` | Primary AWS global region for highest Free Tier coverage. |
| **Operating System** | `Ubuntu 24.04 LTS x86_64` | Official Canonical server image with Linux kernel 6.8. |
| **Block Storage (EBS)** | `20 GB gp3 SSD (Encrypted)` | Encrypted root filesystem ( `/dev/sda1` ) with 3000 IOPS & 125 MB/s baseline. |
| **Memory Extension** | `2.0 GB Swapfile (/swapfile)` | Configured via EC2 UserData bootstrap to prevent OOM during Docker builds. |
| **Networking (VPC)** | `Default VPC (vpc-0e944f63f5c58a72a)` | Standard AWS VPC with public routing table and internet gateway. |
| **Public IPv4** | `34.230.20.223` | Allocated public IPv4 for external HTTPS and SSH management. |
| **Private IPv4** | `172.31.31.238` | Internal VPC subnet address. |
| **SSH Authentication** | `sahayog-hr-key.pem` | RSA 2048-bit key pair registered as `sahayog-hr-key` . |

## 4. AWS Security Group Rules

| Type | Port | Protocol | Source | Security Rationale |
|---|---|---|---|---|
| Inbound | `80` | TCP | `0.0.0.0/0` | Standard HTTP traffic; automatically redirected to HTTPS (443) by Caddy. |
| Inbound | `443` | TCP | `0.0.0.0/0` | Encrypted HTTPS traffic for Web SPA and Mobile App REST API requests. |
| Inbound | `22` | TCP | `0.0.0.0/0` | Administrative SSH shell access authenticated exclusively by PEM private key. |
| Outbound | All | All | `0.0.0.0/0` | Allows container image pulls, Ubuntu security patches, and GitHub sync. |

## 5. Containerized Services Running on EC2

### CADDY 2 PROXY (app-caddy-1)

Acts as the front-facing entry point. Terminates SSL/TLS with zero maintenance using automatic Let's Encrypt certificates. Serves the pre-compiled React frontend SPA directly from the `/srv` mount and proxies API requests to Gunicorn.

### DJANGO BACKEND (app-api-1)

Runs Django 4.2.16 with Gunicorn WSGI workers on Python 3.12-slim. Implements REST APIs, JWT authentication, soft-deletes, role-based access control, attendance rules, and automated payslip PDF generation.

### POSTGRESQL 16 (app-db-1)

Production database using PostgreSQL 16 Alpine. Data is persisted outside the container in Docker volume `app_postgres_data`, guaranteeing zero data loss across container rebuilds or reboots.

### PERSISTENT VOLUMES

- `app_postgres_data` : Relational database records.
- `app_media_data` : Uploaded documents & payslip PDFs.
- `app_caddy_data` : SSL certificates & OCSP state.
- `app_caddy_config` : Caddy dynamic runtime configs.

## 6. Strategic Free-Tier Design Choices

**Why are we not using AWS RDS, AWS S3, or AWS Lambda?**

Managed AWS services introduce monthly recurring costs (e.g. AWS RDS typically costs $15–$30/mo, S3 incurs request charges, and Route 53 incurs zone maintenance fees). By co-locating the database and reverse proxy in lightweight Docker containers on an encrypted EBS volume, Sahayog HR achieves **100% Free-Tier compliance ($0/mo)** while maintaining production-grade reliability and security.

## 7. Live Access & System Endpoints

| Service Endpoint | URL / Location | Access Role |
|---|---|---|
| **Web HRMS Portal** | https://34-230-20-223.sslip.io | Desktop & Mobile Web Users |
| **REST API Base** | https://34-230-20-223.sslip.io/api/v1/ | Mobile App & Frontend Client |
| **API Health Check** | https://34-230-20-223.sslip.io/api/v1/health/ | Public Monitoring / Uptime |
| **Django Admin Console** | https://34-230-20-223.sslip.io/admin/ | Superusers & System Admins |
| **Android APK Release** | app-debug.apk (Direct Download) | Mobile Employees & Admins |

| Service Endpoint | URL / Location | Access Role |
| --- | --- | --- |
| **SSH Shell Command** | `ssh -i deploy/sahayog-hr-key.pem ubuntu@34.230.20.223` | DevOps & Infrastructure Admin |
| **Server Code Path** | `/home/ubuntu/app` | Production Deployment Dir |


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